

Spare parts

In the past, losing an important part of the body due to disease or injury led to death or severe disability. Today, doctors can repair many parts of the body with artificial replacements. In years to come, they may be able to replace organs with fresh new ones grown in a lab from your own body cells. This technology would allow us to live decades longer than today.

REBUILDING THE BODY

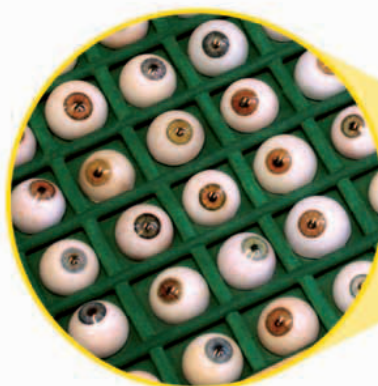
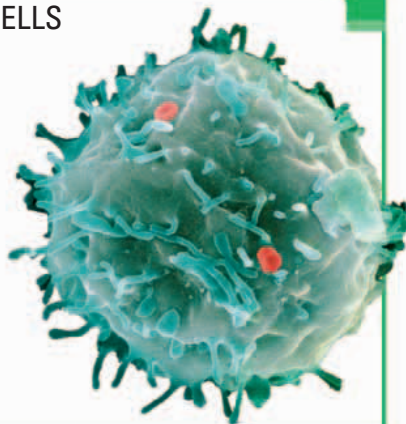
Some of the artificial parts that doctors use to repair damaged bodies are mechanical or electrical devices made of metals and plastics. Others, such as artificial skin or transplanted organs, are made of living tissue. Most of these spare parts do not work as well as the organs they replace.

► **NEW SKIN** *Scientists have discovered how to grow sheets of artificial skin from a person's skin cells. Although they're very thin, these sheets may soon be used to repair damage caused by severe burns.*



STEM CELLS

Some of the cells in the human body have an amazing ability: They can multiply and develop into any kind of tissue, such as skin, blood, or nerves. Such cells are known as stem cells. Scientists are trying to harness this ability of stem cells and create new tissues or organs to repair or replace body parts that are injured or diseased.



▲ **GLASS EYES** *Artificial eyes look realistic but they do not restore vision. However, scientists are developing implants that could restore some form of vision to people who are blind.*

▼ **HIP JOINT** *Damaged by age or disease, some joints, such as the hip, can be repaired with an artificial replacement.*

